

CJ 540

QUANTITATIVE DATA ANALYSIS

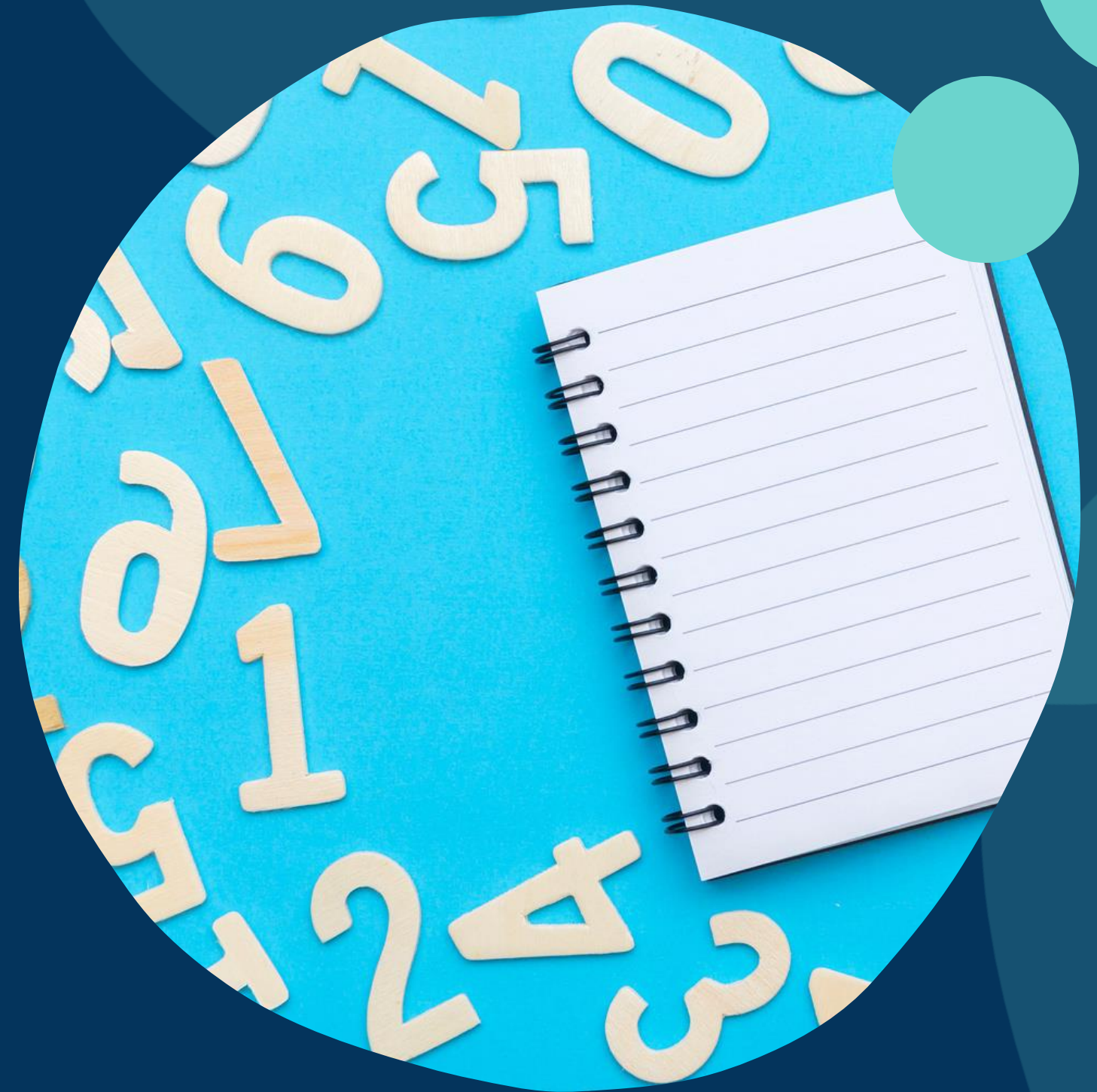
PROFESSOR BURREL VANN JR

QUANTITATIVE DATA

THE NUMERICAL REPRESENTATION OF OBSERVATIONS
FOR THE PURPOSE OF DESCRIBING AND EXPLAINING
PHENOMENA

QUANTITATIVE CODES/CODING

Assigning numbers to attributes of variables



AGE

1 = 1

2 = 2

3 = 3

< 18 = 1

18-24 = 2

25-29 = 3

30+ = 4

SEX

Female = 1

Male = 2

Other = 3

POLITICAL AFFILIATION

Democrat = 1

Republican = 2

Independent = 3

Other = 4

SOCIOECONOMIC STATUS

Working = 1

Middle = 2

Upper-Middle = 3

The One Percent = 4

CODEBOOK

A document that lists the assignments of codes to the attributes composing the variables (meanings of codes, and code labels).

Variable Name	Question Text	Key
ID	Enter the <u>5 digit</u> code given to you by the researcher.	
		3 = Slightly Disagree 4 = Neutral 5 = Slightly Agree 6 = Agree 7 = Strongly Agree
Ext1	I see myself as extraverted, enthusiastic.	1 = Strongly Disagree 2 = Disagree 3 = Slightly Disagree 4 = Neutral 5 = Slightly Agree 6 = Agree 7 = Strongly Agree
Agr1	I see myself as critical, quarrelsome.	1 = Strongly Agree 2 = Agree 3 = Slightly Agree 4 = Neutral 5 = Slightly Disagree 6 = Disagree 7 = Strongly Disagree
Cons1	I see myself as dependable, self-disciplined.	1 = Strongly Disagree 2 = Disagree 3 = Slightly Disagree 4 = Neutral 5 = Slightly Agree 6 = Agree 7 = Strongly Agree
EmSt1	I see myself as anxious, easily upset.	1 = Strongly Agree 2 = Agree 3 = Slightly Agree 4 = Neutral

CODEBOOK EXAMPLE

POLVIEWS

We hear a lot of talk these days about liberals and conservatives. I'm going to show you a seven-point scale on which the political views that people might hold are arranged from extremely liberal—point 1—to extremely conservative—point 7. Where would you place yourself on this scale?

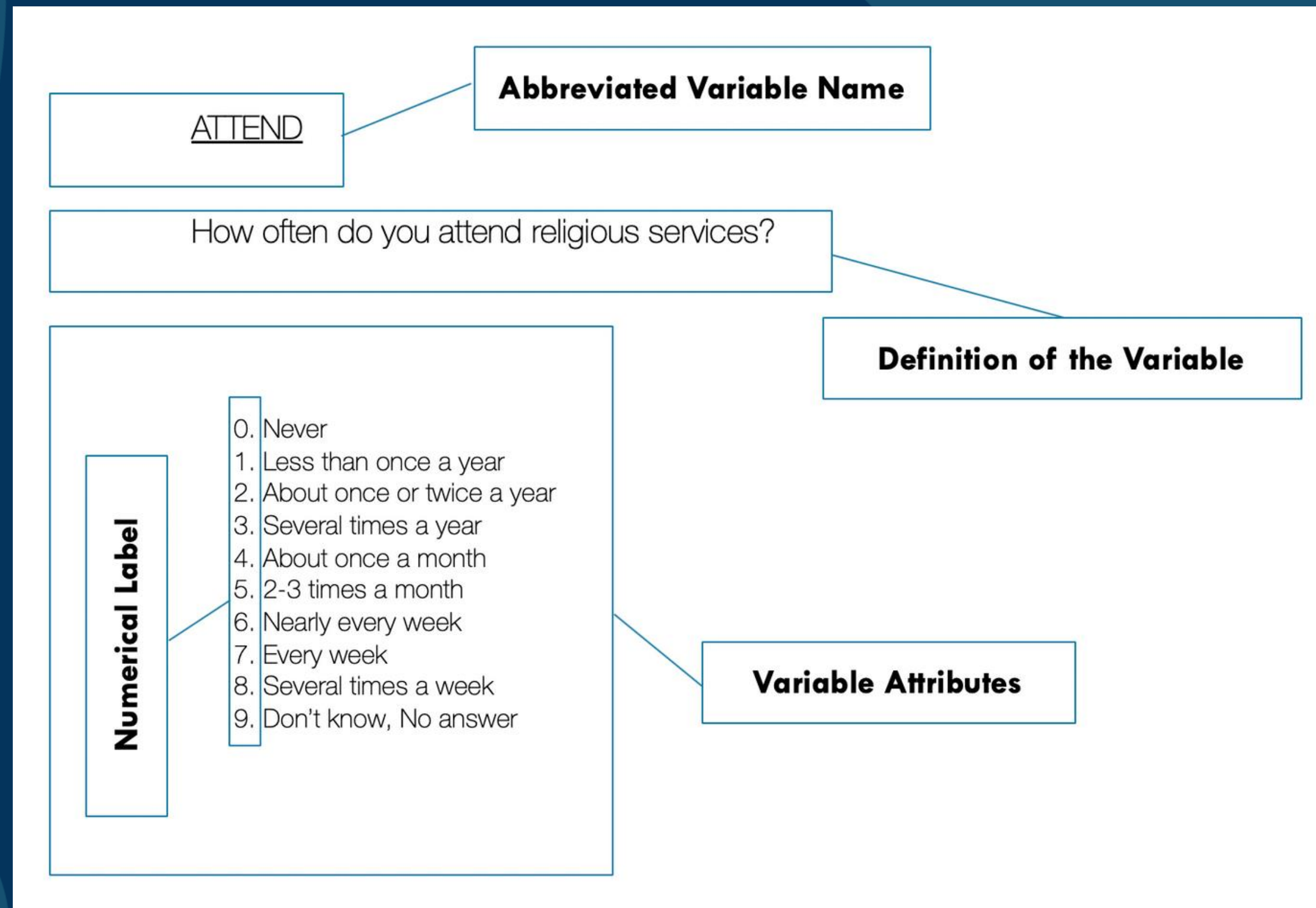
1. Extremely liberal
2. Liberal
3. Slightly liberal
4. Moderate, middle of the road
5. Slightly conservative
6. Conservative
7. Extremely conservative
8. Don't know
9. No answer

ATTEND

How often do you attend religious services?

0. Never
1. Less than once a year
2. About once or twice a year
3. Several times a year
4. About once a month
5. 2–3 times a month
6. Nearly every week
7. Every week
8. Several times a week
9. Don't know, No answer

CODEBOOK EXAMPLE



LEVELS OF MEASUREMENT & QUANTITATIVE CODING

All levels of measurement can be assigned numerical values

NOMINAL

Attributes have only the characteristics of exhaustiveness and mutually exclusiveness. Simple categories.

Attributes that classify, but cannot be ranked

- *religious affiliation, college major, hair color, birthplace, nationality*

ORDINAL

Attributes that classify and can be logically rank-ordered but lack an associated numerical value

- *socioeconomic status, level of conflict, prejudice, conservativeness*

INTERVAL-RATIO

Distance between attributes has meaning.

Attributes classify, can be rank-ordered, and have an equal numerical distance between values

- *temperature (Fahrenheit), IQ score*

INTERVAL-RATIO

AGE

- 1 = 1
- 2 = 2
- 3 = 3

- < 18 = 1
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- 25-29 = 3
- 30+ = 4

NOMINAL

SEX

- Female = 1
- Male = 2
- Other = 3

POLITICAL AFFILIATION

- Democrat = 1
- Republican = 2
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NOMINAL

SOCIOECONOMIC STATUS

- Working = 1
- Middle = 2
- Upper-Middle = 3
- The One Percent = 4

ORDINAL

EXHAUSTIVENESS & MUTUAL EXCLUSIVENESS

Exhaustive:

Ensuring that all possible units can fall into a category
(How do you handle missing responses and if people don't know how to answer)

Mutually Exclusive:

Units can only fall into one category

DATA ENTRY

.CSV (e.g. Excel)

DATA ANALYSIS

R

STATA

SPSS

SAS

TYPES OF QUANTITATIVE METHODS

Univariate Statistics
Bivariate Statistics
Multivariate Statistics

UNIVARIATE STATISTICS

DESCRIBING A SINGLE VARIABLE AT A TIME

UNIVARIATE STATISTICS

- Central Tendency
- Dispersion/Variability
- Frequencies & Graphing

MEASURES OF CENTRAL TENDENCY

MODE

the most frequently observed value or attribute.

Can be used for all variable types: nominal, ordinal, and interval.

MEDIAN

the value of the “middle” case in a rank-ordered set of observations.

Should only be used for two variable types: ordinal, and interval.

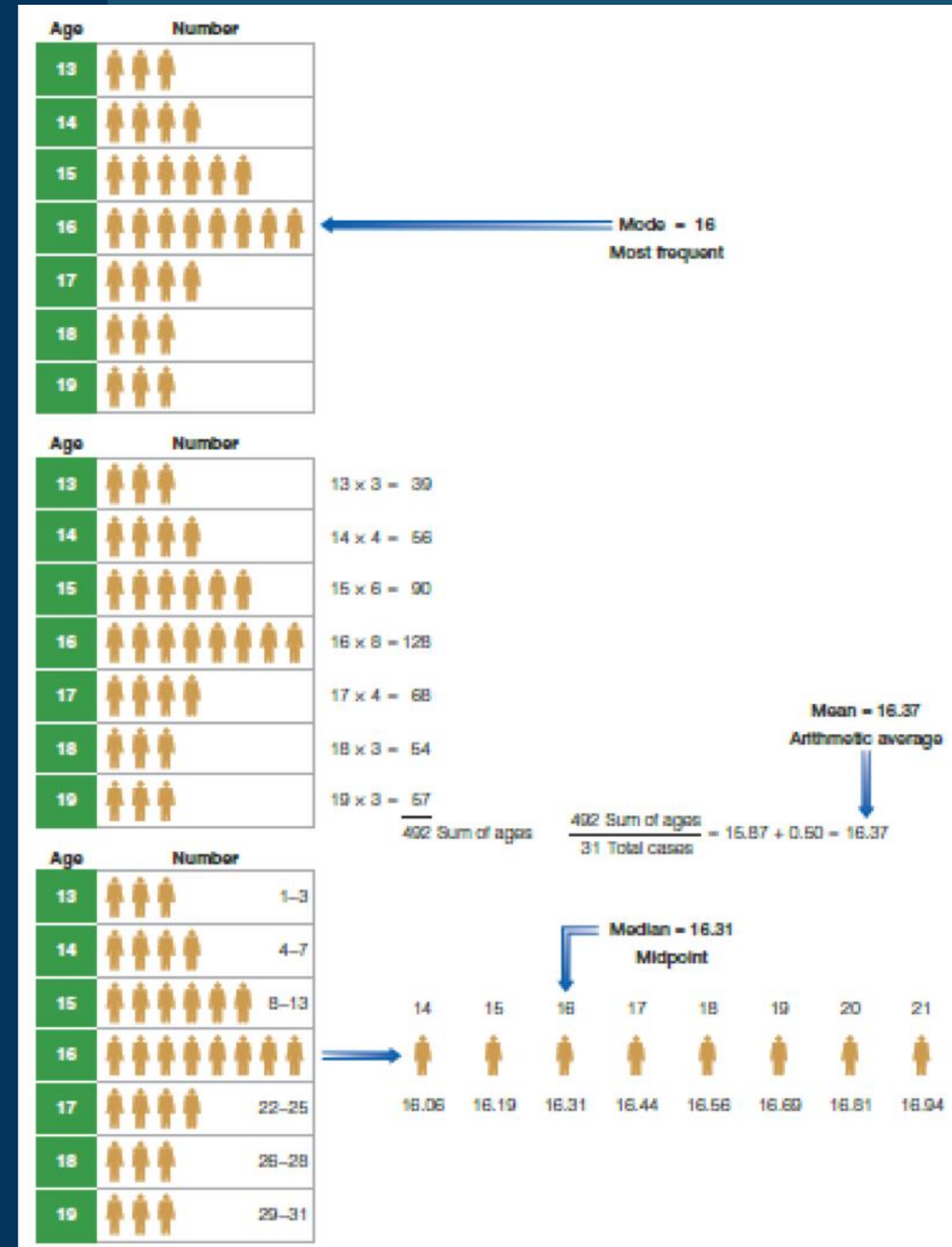
MEAN

computed by summing the values of several observations and dividing by the number of observations.

Average.

Should only be used for interval variables.

UNIVARIATE STATISTICS EXAMPLE



MEASURES OF VARIABILITY (DISPERSION)

RANGE

distance between highest and lowest value

Should only be used for interval variables.

VARIANCE

A measure of dispersion around the mean, but in squared units

Should only be used for interval variables.

STANDARD DEVIATION

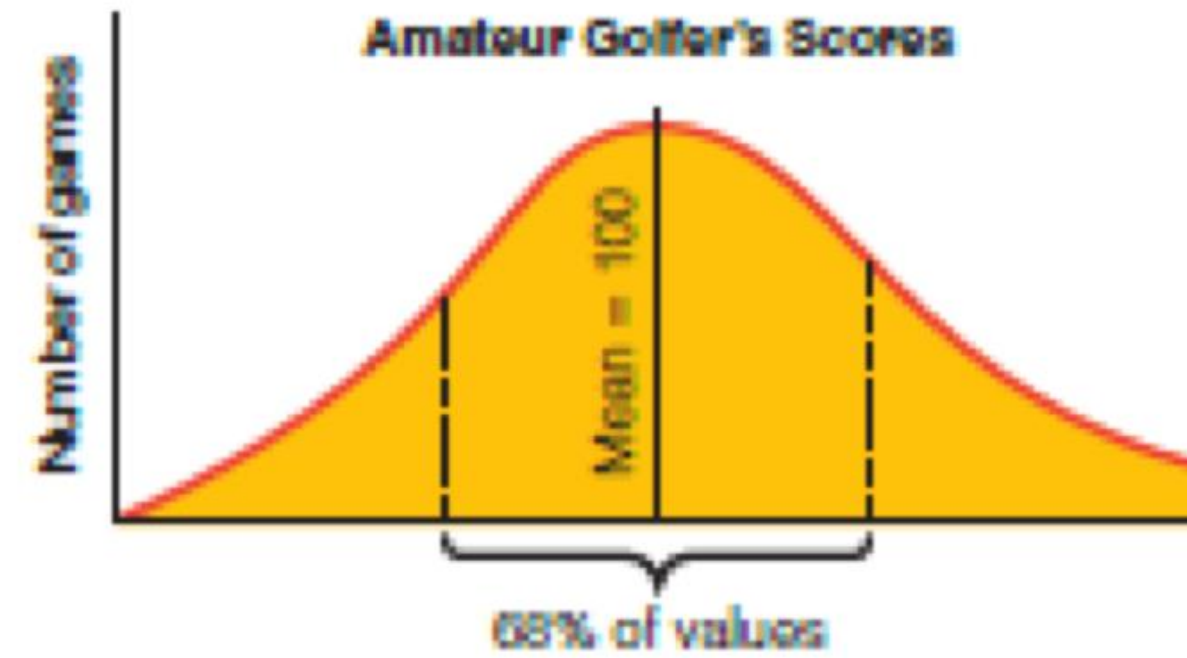
A measure of dispersion around the mean, calculated so that approximately 68 percent of cases fall within one standard deviation of the mean, 95 percent within two, and 99.9 percent within three standard deviations.

Should only be used for interval variables.

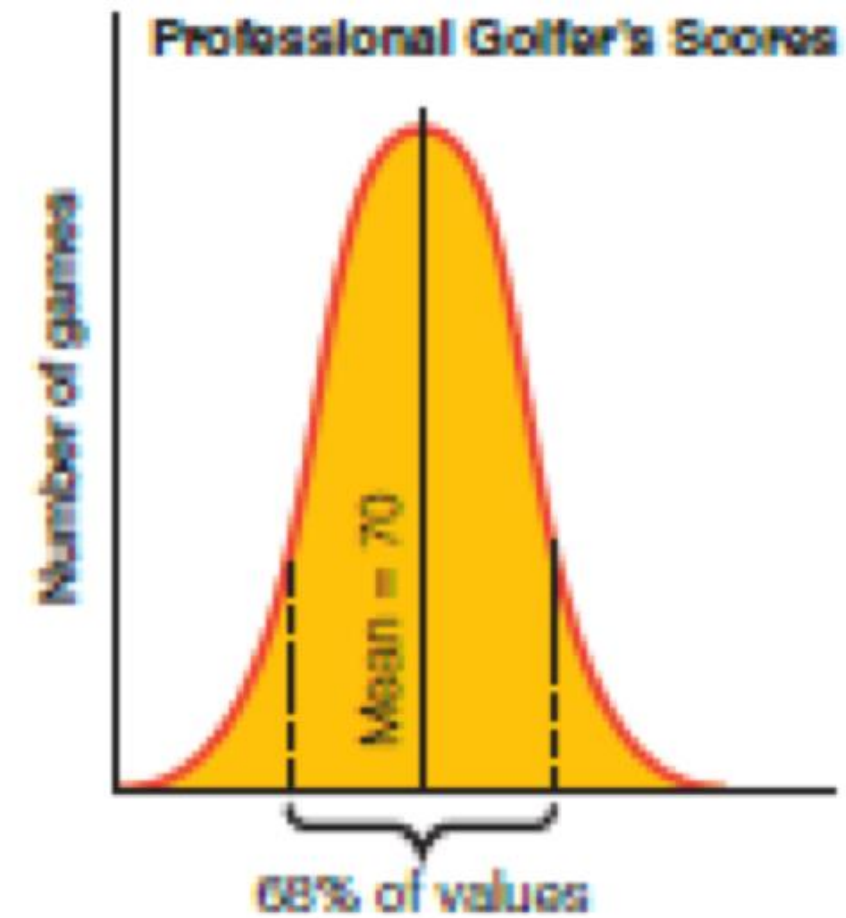
WHY STANDARD DEVIATION?

SD gives us a measure of clustering around the mean (how close, or far, most cases are around the mean).

STANDARD DEVIATION EXAMPLE



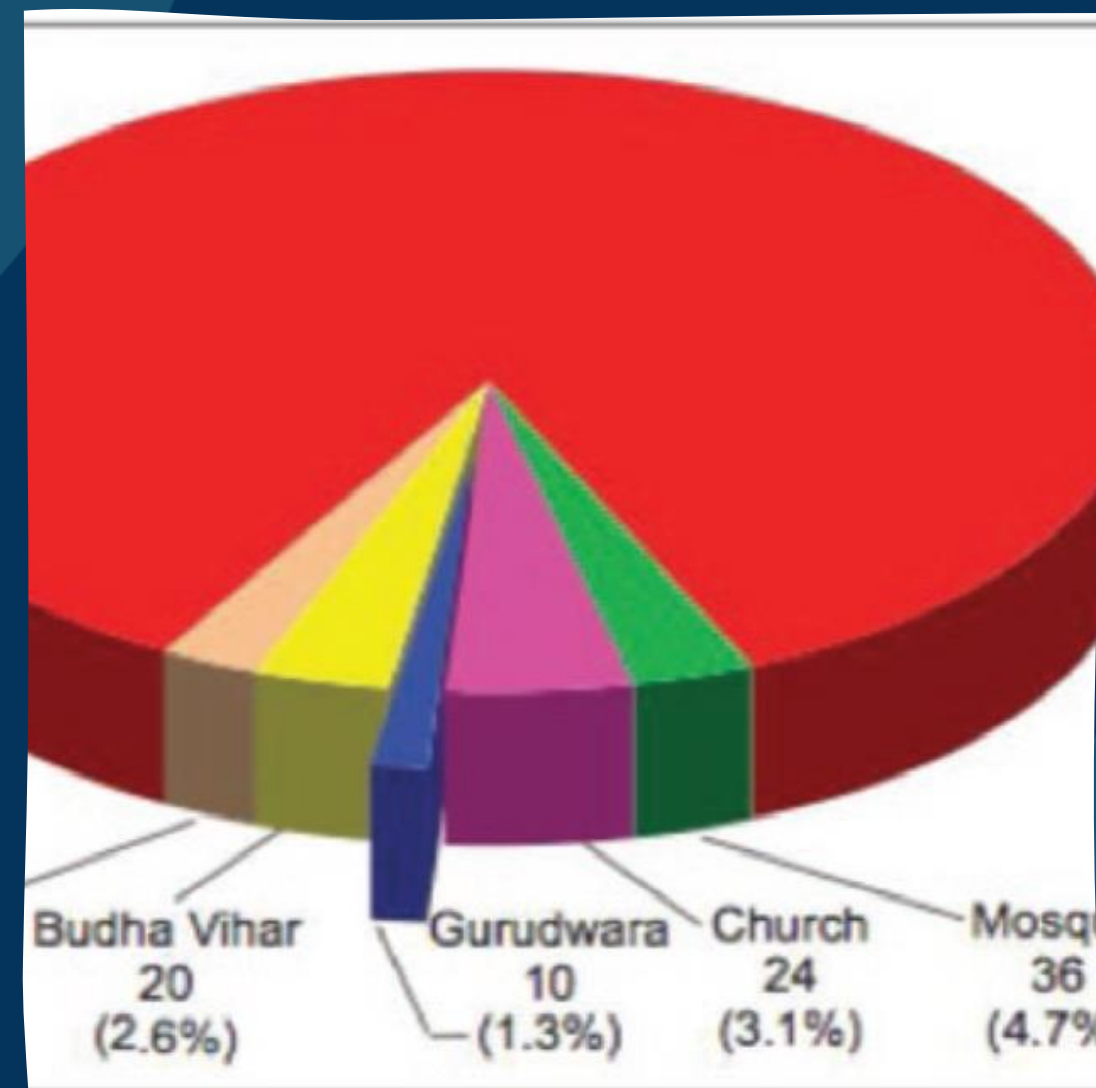
d. Low standard deviation = tightly clustered values



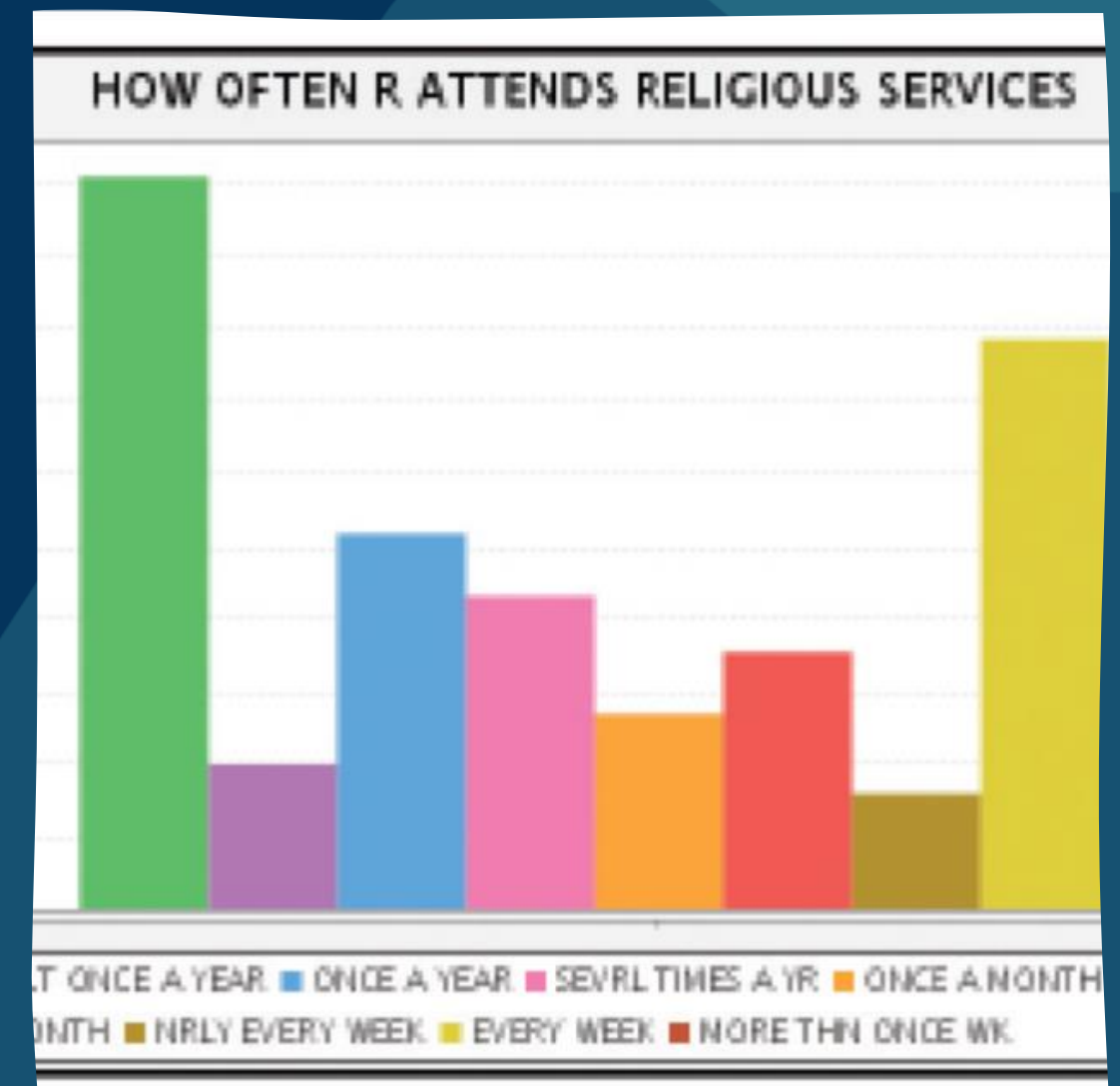
FREQUENCIES & GRAPHING

Frequency Distribution		
Cells contain: -Column percent -Design effect (DEFT) -N of cases		Distribution
0: NEVER	25.2 1.359 496	
1: LT ONCE A YEAR	5.6 1.030 111	
2: ONCE A YEAR	13.4 1.132 264	
3: SEVRL TIMES A YR	9.7 1.098 191	
4: ONCE A MONTH	6.8 1.111 111	

FREQUENCY
DISTRIBUTION



PIE CHART



BAR GRAPH

BIVARIATE STATISTICS

EXAMINING RELATIONSHIPS BETWEEN TWO VARIABLES

BIVARIATE STATISTICS

The analysis of two variables simultaneously, for the purpose of determining an empirical relationship between them.

- Chi Square
- t-Test
- Analysis of Variance (ANOVA)
- Correlation

BIVARIATE STATISTICS

	Dependent Variable	
Independent Variable	Nominal/Ordinal	Interval-Ratio
Nominal/Ordinal	Chi-Square	T-Test or ANOVA
Interval-Ratio	N/A	Correlation

MULTIVARIATE STATISTICS

EXAMINING RELATIONSHIPS BETWEEN THREE OR MORE
VARIABLES

MULTIVARIATE STATISTICS

The analysis of the simultaneous relationships among several variables.

- MANOVA
- Regressions (Multiple/Linear/OLS, Binary Logistic, Multinomial Logistic, Poisson, Negative Binomial, Event History/Survival Analysis, Time-Series)